

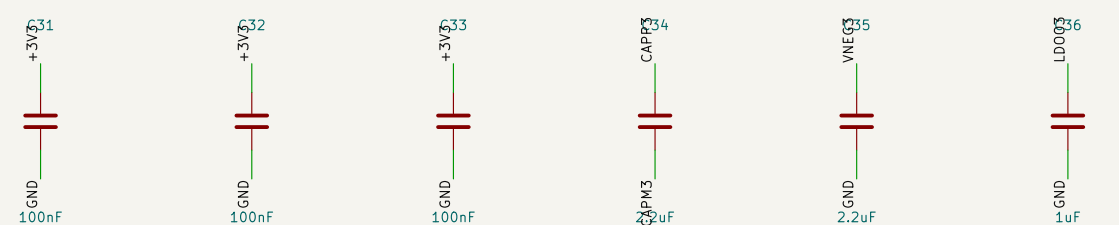
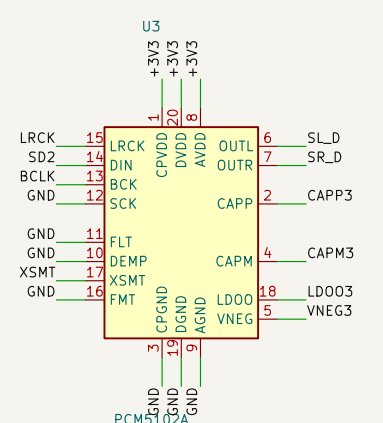
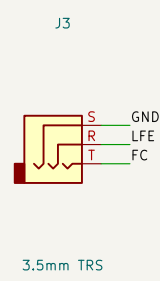
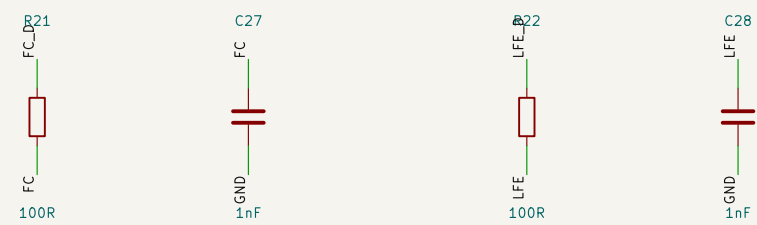
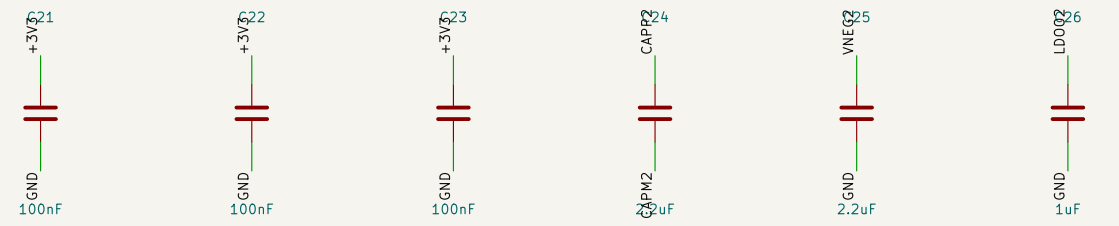
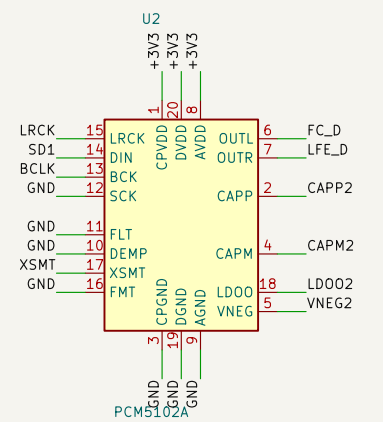
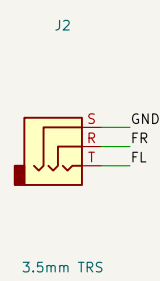
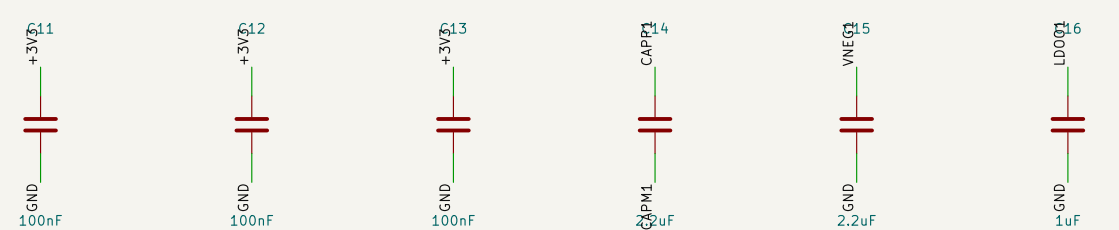
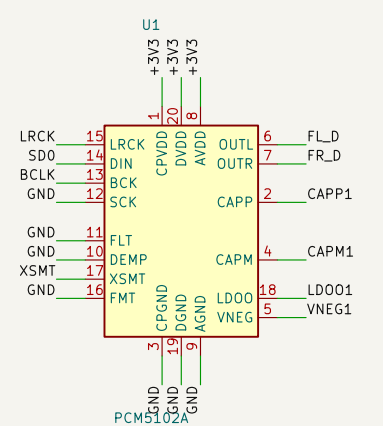
5V from the user port, regulated locally to 3V3.
3x PCM5102A is -90 mA; AP2112K is a 600 mA part.
U4 EN tied to VIN: the rail is always on, muting is
done at the DACs (XSM1), not by cutting power.

R1-R5: series damping at the receiver.

0R is acceptable - a 30 cm run at 3.072 MHz
does not need it. Fitted as cheap insurance.

R6 (4k7 pull-down) is the MUTE INTERLOCK and the reason this is safe to leave plugged in. The FPGA's internal weak pull-up (~25k) cannot overcome it, so any core not driving EN push-pull leaves the DACs muted. Plugged into a real USB3 port, contact 7 is ground – also muted. R7/C5 de-glitch the level.

R8 is NOT FITTED: see the note next to it before assuming contact 7 reaches this board as a signal.



3.5mm TRS

PCMS102A output is GROUND-CENTRED – the internal charge pump (C.4 flying cap across CAPP/CAPM, C.5 on VNEG) generates the negative rail – so NO DC-blocking capacitors are needed or wanted. The output R/C is RF rejection and short-circuit protection only, not a reconstruction filter.

SOCK tied to GND selects BCK-PLL mode, so no MCLK wire is needed and the user port has a contact to spare for EN. Channel order is the PC analog convention, so a stock 3x3.5mm-to-6xRCA cable maps straight through. An HDMI sink wants LFE/C in the other order – remapped in the transmitter.